

An Intersemiotic Analysis Of Language And Gesture In Early Years Classroom Discourse In North-east Rivers State, Nigeria

By

Dr. Joseph Ahaotu

joseph.ahaotu@uniport.edu.ng

Department Of English Studies, Faculty Of Humanities,
University Of Port Harcourt, Rivers State

And

Nwogu Hope Ozioma

nwoguhope2013453@gmail.com

Department Of English And Communication Art, Faculty Of Humanities,
Ignatius Ajuru University Of Education
Port Harcourt, Rivers State

ABSTRACT

Language learning in early childhood classrooms is inherently multimodal, involving the coordinated use of speech, gesture, facial expression, gaze, body movement and other semiotic resources that jointly construct meaning. Although previous studies have examined classroom interaction and gesture in language learning, limited attention has been paid to how these resources function intersemiotically within Nigerian early years classrooms using the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework. This study therefore investigates the intersemiotic relationship between language and gesture in selected early years classrooms in North-East Rivers State, Nigeria. Specifically, it examines the communicative functions of language and gesture, analyses how their interaction constructs ideational, interpersonal and textual meanings, and explores their contribution to English language learning among young learners. The study adopted a qualitative descriptive research design. Data were collected through classroom observations, classroom photographs and video recordings from three purposively selected nursery schools and analysed using the analytical principles of Systemic Functional Multimodal Discourse Analysis. The findings reveal that language and gesture operate as complementary semiotic resources that enhance meaning-making, sustain learner engagement, facilitate comprehension and improve classroom participation. The analysis further demonstrates that gestures reinforce spoken language through intersemiotic complementarity by simultaneously realising ideational, interpersonal and textual metafunctions during classroom interaction. The study contributes to the growing body of multimodal discourse scholarship by demonstrating the applicability of SF-MDA to Nigerian early childhood classroom discourse and provides practical implications for language pedagogy, teacher education and multimodal instructional practices.

Keywords: *Intersemiotic analysis; gesture; classroom discourse; early years education; Systemic Functional Multimodal Discourse Analysis; English language learning; Nigeria.*

INTRODUCTION

Language is the principal medium through which human beings communicate, construct knowledge, and participate in social life. It extends beyond spoken and written words to include a wide range of semiotic resources such as gestures, facial expressions, gaze, posture, body movement, images and other visual signs that collectively facilitate meaning-making. From a

social semiotic perspective, communication is inherently multimodal because meaning is negotiated through the interaction of multiple semiotic resources rather than language alone. In educational settings, particularly during the early years of schooling, these multimodal resources are indispensable in supporting children's cognitive, linguistic and social development.

Early childhood represents one of the most critical periods of language acquisition. During this stage, children rapidly develop their vocabulary, pronunciation, comprehension and communicative competence through continuous interaction with teachers, peers and their immediate environment. Vygotsky's (1978) sociocultural theory emphasises that learning is socially mediated and that children's language development occurs primarily through meaningful interaction with more knowledgeable others. Consequently, classroom communication extends beyond verbal instruction to include gestures, facial expressions, demonstrations, visual representations and other embodied forms of communication that enable young learners to construct meaning more effectively.

Classroom discourse refers to the patterns of communication that occur between teachers and learners as well as among learners during instructional activities. It encompasses spoken language, questioning techniques, feedback, explanations, demonstrations, gestures, gaze, classroom artefacts and other communicative resources employed during teaching and learning. Hall and Walsh (2002) observe that classroom discourse constitutes the foundation of effective teaching because it shapes learners' participation, comprehension and overall learning outcomes. Similarly, Kumaravadivelu (1999) argues that what occurs during classroom interaction largely determines the extent to which educational objectives are achieved. Consequently, understanding the multimodal nature of classroom discourse has become increasingly important in language education research.

Among the various multimodal resources employed in classroom interaction, gesture occupies a particularly significant position. Gestures are meaningful bodily movements that accompany or sometimes replace spoken language during communication. Kendon (2004) describes gesture as an integral component of human communication rather than a mere accompaniment to speech, while McNeill (1992) argues that speech and gesture form a unified communicative system that jointly expresses meaning. In early years classrooms, teachers frequently employ pointing gestures, iconic gestures, facial expressions, body movements and demonstrations to explain unfamiliar vocabulary, illustrate abstract concepts, direct learners' attention and reinforce spoken instructions. Young learners equally rely on gestures to express understanding, respond to questions and negotiate meaning even when their verbal abilities are still developing.

Recent scholarship in multimodal discourse studies has shifted attention from viewing language as an isolated system to understanding communication as the coordinated deployment of multiple semiotic resources. Studies by O'Halloran (2008), Lim (2017), Jewitt (2016), Norris (2020) and other multimodal scholars demonstrate that meaning emerges through the interaction of speech, gesture, images, gaze, movement and spatial organisation. This interrelationship, referred to as intersemiosis, explains how different semiotic resources complement one another to produce meanings that cannot be achieved by any single mode independently. In classroom discourse, intersemiosis enables teachers to simplify complex concepts, maintain learners' attention, encourage participation and enhance language

comprehension.

Systemic Functional Multimodal Discourse Analysis (SF-MDA), developed from Halliday's Systemic Functional Linguistics and subsequently expanded by O'Halloran, Lim and other multimodal scholars, provides an appropriate analytical framework for investigating how multiple semiotic resources function together during communication. The framework analyses meaning through three metafunctions—ideational, interpersonal and textual—which explain how language, gesture and other semiotic resources simultaneously represent experience, establish social relationships and organise coherent discourse. Unlike traditional discourse analysis, SF-MDA enables researchers to examine not only spoken language but also the contribution of gestures, gaze, posture, movement and visual resources to classroom meaning-making.

Despite increasing international scholarship on multimodal classroom interaction, empirical studies applying SF-MDA to early years classroom discourse within the Nigerian context remain limited. Existing Nigerian studies have focused predominantly on verbal classroom interaction, teacher talk or general language pedagogy, with comparatively little attention given to the intersemiotic relationship between language and gesture in early childhood English language classrooms. Consequently, there is insufficient understanding of how teachers and pupils coordinate verbal and non-verbal semiotic resources to facilitate language learning in Nigerian early years education.

It is against this background that the present study investigates the intersemiotic relationship between language and gesture in selected early years classrooms in North-East Rivers State, Nigeria. Drawing on the analytical principles of Systemic Functional Multimodal Discourse Analysis, the study examines how verbal and non-verbal semiotic resources interact to construct ideational, interpersonal and textual meanings during classroom discourse. In doing so, the study contributes to multimodal discourse scholarship by extending the application of SF-MDA to Nigerian early childhood education and provides insights that can inform classroom pedagogy, teacher education and language teaching practices.

Language and gesture function together as complementary semiotic resources in early years classroom discourse, playing a crucial role in facilitating English language learning. While previous studies have examined classroom interaction, gesture, and multimodal communication, limited attention has been given to how language and gesture interact intersemiotically to construct meaning in Nigerian early years classrooms using the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework.

This gap has limited understanding of how verbal and non-verbal semiotic resources jointly realise ideational, interpersonal, and textual meanings during classroom interaction. Consequently, there is insufficient empirical evidence to explain how these multimodal resources enhance learners' comprehension, participation, and language development in early childhood education.

This study addresses this gap by investigating the intersemiotic relationship between language and gesture in selected early years classrooms in North-East Rivers State, Nigeria, using SF-MDA. The study contributes to multimodal discourse scholarship by demonstrating how verbal

and non-verbal semiotic resources complement one another to enhance meaning-making and English language learning.

The aim of this study is to investigate the intersemiotic relationship between language and gesture in early years classroom discourse in selected schools in North-East Rivers State, Nigeria, using the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework.

The specific objectives of the study are to: examine how language and gesture function as complementary semiotic resources in early years classroom discourse;

analyse how language and gesture jointly realise the ideational, interpersonal, and textual metafunctions in classroom interaction;

investigate how the intersemiotic relationship between language and gesture facilitates English language learning among early years learners; and

determine the pedagogical implications of language–gesture interaction for effective English language teaching in early years classrooms.

The following research questions guided the study:

- How do language and gesture work together in early years classroom discourse?
- How do language and gesture realise the ideational, interpersonal, and textual metafunctions in classroom interaction?
- How does language–gesture interaction support English language learning among early years learners?
- What are the pedagogical implications of language–gesture interaction for effective English language teaching in early years classrooms?

LITERATURE REVIEW

Classroom discourse has increasingly been recognised as a multimodal communicative process in which meaning is constructed through the interaction of language with other semiotic resources such as gesture, gaze, facial expression, posture, and body movement. From a social semiotic perspective, communication extends beyond spoken language to include multiple modes that operate simultaneously to facilitate meaning-making. Consequently, classroom interaction, particularly in early years education, is best understood as a coordinated use of verbal and non-verbal resources.

Kendon (2004) argue that gesture is an integral component of communication rather than a mere accompaniment to speech. Similarly, Jewitt (2016), Lim (2019), and Norris (2020) maintain that multimodal resources function collaboratively to enhance meaning, classroom participation, and learner engagement. Within early years classrooms, teachers employ gestures to reinforce spoken language, direct learners' attention, illustrate concepts, provide feedback, and facilitate vocabulary acquisition. Young learners also use gestures to express understanding, negotiate meaning, and participate actively in classroom interaction.

Recent studies on multimodal classroom discourse have demonstrated that the integration of language and gesture enhances comprehension, improves learner participation, and promotes effective English language learning. However, much of the existing literature has focused on teacher talk, gesture as an isolated mode, or multimodal communication in general. Comparatively few studies have examined the intersemiotic relationship between language and gesture within Nigerian early years classrooms using the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework.

The present study addresses this gap by investigating how language and gesture operate as complementary semiotic resources in selected early years classrooms in North-East Rivers State, Nigeria. By applying the SF-MDA framework, the study explains how these resources jointly realise ideational, interpersonal, and textual meanings during classroom discourse and contribute to effective English language learning.

THEORETICAL FRAMEWORK

This study is anchored on Systemic Functional Multimodal Discourse Analysis (SF-MDA), an extension of Halliday's (1978, 1985) Systemic Functional Linguistics developed by O'Halloran (2008), Lim (2019), and other multimodal discourse scholars. SF-MDA provides a comprehensive framework for examining how multiple semiotic resources—including language, gesture, gaze, facial expression, posture, and images—interact to construct meaning within a communicative event.

Three metafunctions of language: the ideational metafunction, which represents experience and ideas; the interpersonal metafunction, which establishes social relationships and interaction between participants; and the textual metafunction, which organises communicative resources into a coherent message. SF-MDA extends these metafunctions beyond verbal language to explain how different semiotic modes work together to produce meaning.

A central concept in SF-MDA is intersemiosis, which refers to the complementary relationship among semiotic resources in meaning-making. Rather than functioning independently, language and gesture operate simultaneously to reinforce, expand, or clarify one another during communication. In classroom discourse, teachers frequently combine spoken explanations with gestures, facial expressions, gaze, and body movements to facilitate learners' understanding, sustain attention, and encourage active participation.

The choice of SF-MDA is appropriate for this study because it enables a systematic analysis of the intersemiotic relationship between language and gesture in early years classroom discourse. The framework provides the analytical tools for explaining how these semiotic resources jointly realise ideational, interpersonal, and textual meanings during English language instruction. It therefore aligns with the aim, objectives, and research questions of the study and offers a robust basis for interpreting the multimodal interactions observed in the selected classrooms.

METHODOLOGY

This study adopted a qualitative descriptive research design to investigate the intersemiotic relationship between language and gesture in early years classroom discourse. The qualitative approach was considered appropriate because it provides an in-depth understanding of naturally

occurring classroom interactions and the meanings constructed through verbal and non-verbal semiotic resources.

The study was conducted in three purposively selected early years schools in North-East Rivers State, Nigeria. Participants comprised English language teachers and early years learners whose classroom interactions constituted the primary data for the study. Purposive sampling was employed to select classrooms where multimodal instructional practices were evident.

Data were collected through non-participant classroom observations, video recordings, and classroom photographs. These instruments enabled the researcher to capture naturally occurring verbal and non-verbal interactions during English language lessons. Ethical approval was obtained from the school authorities, and informed consent was sought from the participating teachers. To ensure confidentiality, the identities of the schools, teachers, and learners were anonymised.

The data were analysed using the Systemic Functional Multimodal Discourse Analysis (SFMDA) framework. Classroom interactions were coded and analysed according to the ideational, interpersonal, and textual metafunctions, with particular attention to the intersemiotic relationship between language and gesture in meaning-making. The findings were interpreted qualitatively and presented through selected classroom extracts and images to illustrate how verbal and non-verbal semiotic resources jointly facilitated English language learning

DATA ANALYSIS AND DISCUSSION OF FINDINGS

Classroom Episode 1 of DOPEC'S School: Teacher's Explanation Accompanied by Gestures



During an English language lesson on parts of the body, the teacher introduced the vocabulary head, eyes, nose, and mouth. As each word was pronounced, the teacher simultaneously pointed to the corresponding body part while maintaining eye contact with the learners. The pupils responded by repeating the words and imitating the teacher's gestures.

From the ideational perspective, the teacher's speech identified the body parts, while the accompanying pointing gestures visually represented the same concepts. The combination of verbal and gestural modes enabled learners to associate the spoken words with their physical referents, thereby facilitating vocabulary development and comprehension.

From the ideational perspective, the teacher's speech identified the body parts, while the accompanying pointing gestures visually represented the same concepts. The combination of

verbal and gestural modes enabled learners to associate the spoken words with their physical referents, thereby facilitating vocabulary development and comprehension.

From the interpersonal perspective, the teacher's eye contact, facial expression, and encouraging gestures created a supportive learning environment that promoted learners' participation. The pupils' imitation of both speech and gesture demonstrated active engagement and collaborative meaning-making during classroom interaction.

From the textual perspective, speech and gesture operated cohesively to organise the lesson into a meaningful sequence. Each gesture coincided with the spoken word, creating a coherent multimodal message that enhanced learners' understanding and sustained their attention throughout the activity.

The analysis demonstrates an intersemiotic relationship between language and gesture, as both semiotic resources complemented one another in constructing meaning. Rather than functioning independently, the teacher's gestures reinforced the spoken language, making vocabulary learning more meaningful and interactive. This finding supports the principles of Systemic Functional Multimodal Discourse Analysis, which views meaning as the product of multiple interacting semiotic resources.

Classroom Episode 2 of NOBSAM School : Teacher's Demonstration of Action Verbs through Gesture



The teacher introduced action verbs such as clap, jump, stand, and sit during an English language lesson. As each verb was pronounced, the teacher performed the corresponding action and instructed the learners to imitate the movements. The pupils responded by carrying out the actions while repeating the words aloud.

From the ideational perspective, the teacher's gestures visually represented the meanings of the action verbs, while the spoken language provided their verbal labels. The simultaneous use of speech and body movement enabled learners to associate the lexical items with the corresponding physical actions, thereby enhancing comprehension and vocabulary acquisition.

From the interpersonal perspective, the teacher's demonstrations, encouraging facial expressions, and interactive questioning promoted learners' active participation. The pupils' imitation of both the spoken words and gestures reflected engagement, collaboration, and shared meaning-making within the classroom.

From the textual perspective, the teacher coordinated speech and gesture in a logical sequence that ensured coherence throughout the lesson. Each verbal instruction was immediately reinforced by a corresponding gesture, creating a unified multimodal message that maintained learners' attention and facilitated understanding.

The analysis reveals that language and gesture functioned intersemiotically to achieve effective classroom communication. The gestures did not merely accompany speech but complemented and reinforced it, enabling learners to understand and perform the action verbs with greater accuracy. This finding supports the principles of Systemic Functional Multimodal Discourse Analysis, which emphasises that meaning emerges through the integration of multiple semiotic resources during classroom discourse.

Classroom Episode 3 of Holy Seed school: Teacher–Learner Interaction during a Classroom Song



The teacher introduced an English action song, Head, Shoulders, Knees and Toes, to reinforce learners' understanding of body-part vocabulary. As the song was sung, the teacher pointed to each body part while performing the corresponding actions. The learners actively participated by singing, imitating the gestures, and pointing to the named body parts.

From the ideational perspective, the spoken words identified the body parts, while the accompanying gestures provided their visual and physical representations. The integration of speech, movement, and rhythm enabled learners to associate the vocabulary with concrete actions, thereby strengthening comprehension and vocabulary retention.

From the interpersonal perspective, the teacher's expressive gestures, smiling facial expressions, eye contact, and enthusiastic delivery encouraged learners' active participation and sustained classroom engagement. The learners' coordinated singing and gestures reflected shared attention, cooperation, and effective teacher–learner interaction throughout the lesson.

From the textual perspective, the synchronisation of language, music, rhythm, and gesture created a coherent multimodal learning experience. The orderly sequence of verbal and non-verbal resources ensured that each stage of the lesson complemented the next, making the classroom discourse meaningful, organised, and easy for the learners to follow.

Overall, the analysis demonstrates that language, gesture, and song functioned intersemiotically to enhance meaning-making in the classroom. Rather than operating independently, these semiotic resources complemented one another in realising the ideational, interpersonal, and textual metafunctions. This finding further confirms the relevance of the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework in explaining how multimodal classroom discourse promotes effective English language learning among early years learners.

SUMMARY OF THE ANALYSIS

The findings reveal that language and gesture function intersemiotically to facilitate meaning-

making in early years classroom discourse. Across the classroom episodes, speech was consistently complemented by gestures, enabling learners to understand concepts more effectively and participate actively in classroom interaction. The analysis further shows that language and gesture jointly realised the ideational, interpersonal, and textual metafunctions, thereby enhancing English language learning. These findings affirm the relevance of the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework in explaining multimodal meaning-making in early years classrooms.

DISCUSSION OF FINDINGS

The findings demonstrate that language and gesture function as complementary semiotic resources in early years classroom discourse. Teachers consistently integrated speech with gestures to clarify meaning, sustain learners' attention, and encourage active participation. This supports McNeill's (1992) assertion that speech and gesture form a unified communicative system and Kendon's (2004) view that gestures are integral to human communication rather than mere accompaniments to speech.

The study further revealed that language and gesture jointly realised the ideational, interpersonal, and textual metafunctions of the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework. This finding is consistent with O'Halloran (2008) and Jewitt (2016), who argue that meaning is constructed through the interaction of multiple semiotic resources rather than language alone. It also corroborates Lim's (2019) position that multimodal resources enhance classroom interaction and learning.

The study contributes to existing scholarship by providing empirical evidence from Nigerian early years classrooms on how language and gesture operate intersemiotically to facilitate English language learning. It therefore extends the application of the SF-MDA framework to classroom discourse in the Nigerian context.

CONCLUSION

This study examined the intersemiotic relationship between language and gesture in early years classroom discourse using the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework. The findings show that language and gesture operate collaboratively to construct meaning, enhance learners' participation, and support English language learning. The study concludes that effective classroom communication depends on the coordinated use of verbal and non-verbal semiotic resources and recommends the integration of multimodal instructional practices in early years English language teaching.

RECOMMENDATIONS

Based on the findings of this study, the following recommendations are made:

Early years teachers should deliberately integrate gestures with spoken language to enhance learners' comprehension and participation during English language instruction.

Teacher education programmes should incorporate training in multimodal pedagogical strategies to improve classroom communication and learner engagement.

Curriculum developers should promote the use of multimodal instructional resources that encourage the coordinated use of verbal and non-verbal semiotic modes in English language teaching.

Further studies should extend the application of the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework to other educational contexts and levels of language learning in Nigeria.

Contribution to Knowledge

This study contributes to multimodal discourse scholarship by demonstrating how language and gesture function intersemiotically to construct meaning in early years classroom discourse. It extends the application of the Systemic Functional Multimodal Discourse Analysis (SF-MDA) framework to the Nigerian context and provides empirical evidence on the pedagogical value of integrating verbal and non-verbal semiotic resources in English language teaching. The findings also offer practical insights for teachers, teacher educators, and curriculum developers seeking to improve classroom interaction and language learning among early years learners.

REFERENCE

- Erfanian M.J, Elahi S.M, Akbari O. (2019). Systemic functional multinational discourse analysis of teaching students development class – Sir Room materials. *Teach.education* 24 1 – 23.
- Halliday, M. A. K. (1978). *Language as Social Semiotic: The social interpretation of language and meaning*.
- Jewitt, C. (2016). *The routledge handbook of multimodal analysis (2nd ed.)*.
- Kendon, A. (2004). *Gesture: visible action as utterance*.
- Lim, F. V. (2019). *Developing a systemic functional approach to multimodal discourse analysis*.
- Lim, F.V. (2004) Developing an integrative multisemiotic model. In: O'Halloran, K.L. (ed) *Multimodal discourse analysis: systemic-functional perspectives*. *Continuum*, 220-246.
- Matthiessen, C.I.M.M. (2009). Multisemiosis and context-based register typology: Registeral variation - in the complementarity of semiotic systems. In: Ventola, E and Guijarro, A.J.M (eds) *The world told and the world shown: Multisemiotic issues*. *Palgrave Macmillan: Hampshire*, 11- 38.
- McNeill, D. (1992). *Hand and mind: What gestures reveal about thought*.
- Norris, S. (2004). *Analyzing multimodal interaction: A methodological framework*.
- Norris, S. (2020). *Multimodal discourse analysis: A conceptual framework*.
- O'Halloran, K. L. (2008). *Systemic functional-multimodal discourse analysis (SF-MDA)*.
- O'Halloran, K.L. (2005). *Mathematical discourse: Language, symbolism and visual images*.
- Peng J.E, Zhan, L.Chen Y. (2017). The mediation of multinational affordances on willingness to communicate in the english as a foreign language classroom . *TESOL* a.51302 – 33,10.1002.
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*.